AQA 3.2.4 Properties of Period 3 elements and their oxides

AQA A-level Chemistry · Revision questions · Sheet 2 of 6

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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19 marks · about 25 minutes

Key information

- $Mg + H_2O(g) \rightarrow MgO + H_2$ (steam); $2Na + 2H_2O \rightarrow 2NaOH + H_2$
- Basic oxides react with acids to give salt + water; acidic oxides react with bases (e.g. $SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$)
- Na_2O and MgO are giant ionic and basic: $Na_2O + H_2O \rightarrow 2NaOH$ (pH 12–14); $MgO + H_2O \rightarrow Mg(OH)_2$ (pH 9–10)
- Al_2O_3 is insoluble and amphoteric; SiO_2 is insoluble, giant covalent and acidic (reacts with hot conc NaOH)
- P_4O_{10}, SO_2, SO_3 are simple molecular and acidic: $P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$; $SO_2 \rightarrow H_2SO_3$; $SO_3 \rightarrow H_2SO_4$

Section A: Short-answer questions

01 Give an equation for the reaction of sodium oxide with hydrochloric acid.

[1 mark]

02 Give an equation, including state symbols, for the reaction of magnesium with steam. State **two** observations for this reaction.

[3 marks]

03 Give an equation for the reaction of magnesium oxide with water. Suggest the pH of the solution formed.

[2 marks]

04 Equal amounts of three Period 3 elements are added to separate 100 cm^3 samples of water. Which list is correct when the solutions formed from these elements are arranged in order of decreasing pH?

[1 mark]

A chlorine, magnesium, sodium
B chlorine, sodium, magnesium
C magnesium, sodium, chlorine
D sodium, magnesium, chlorine

05 Aluminium oxide is described as *amphoteric*. State the meaning of this term.

[1 mark]

06 Which is **not** a correct statement about magnesium hydroxide?

- It is used to neutralise stomach acid
- It forms a solution with pH = 14 at 25 °C
- It has the empirical formula H_2MgO_2

[1 mark]

Section B: Longer questions

07 This question is about Period 3 elements and their compounds.

07.1 Give an equation for the reaction of aluminium oxide with sulfuric acid.

[1 mark]

07.2 Identify a reagent or test that could be used to distinguish between aqueous solutions of sulfur dioxide and sulfur trioxide with the same concentrations. State the observation in each case.

[3 marks]

08 This question is about Period 3 elements and their oxides.

08.1 Give an equation for the reaction between phosphorus and an excess of oxygen.

[1 mark]

08.2 Give an equation for the reaction between sulfur dioxide and water.

[1 mark]

08.3 Give the displayed formula for the anion formed when sulfur trioxide reacts with water.

[1 mark]

08.4 Give an equation for the reaction of magnesium with steam. State one observation made.

[2 marks]

08.5 Give an equation to show how an excess of magnesium oxide reacts with phosphoric acid (H_3PO_4).

[1 mark]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	$Na_2O + 2HCl \rightarrow 2NaCl + H_2O$	*Source: Practice question (written for this set)*	1
02	$Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$ bright / white flame / light white / grey ash / powder	*Source: 7404/1 Jun 2023 Q02.3* State symbols essential Do not allow ppt	3
03	$MgO + H_2O \rightarrow Mg(OH)_2$ pH 9–10	*Source: Practice question (written for this set)* Allow pH 8–10	2
04	**D**	*Source: 7405/3 Jun 2025 Q30*	1
05	Reacts with both acids and bases	*Source: Practice question (written for this set)*	1
06	It forms a solution with pH = 14 at 25 °C	*Source: 7405/1 Jun 2022 Q05.1*	1
07.1	$Al_2O_3 + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H_2O$	*Source: 7405/1 Jun 2022 Q05.2–05.3* Allow multiples; ignore state symbols	1
07.2	Universal indicator: $SO_2(aq)$ orange-red, $SO_3(aq)$ red **or** pH meter: $SO_2(aq)$ pH 2–3, $SO_3(aq)$ pH 0–1 **or** metal carbonate / Mg: $SO_2(aq)$ slower effervescence, $SO_3(aq)$ faster **or** acidified $KMnO_4$ / $K_2Cr_2O_7$: $SO_2(aq)$ colour change, $SO_3(aq)$ no visible change **or** acidified $BaCl_2$: $SO_2(aq)$ no visible change, $SO_3(aq)$ white precipitate	If reagent is incomplete lose M1 and mark on	3
08.1	$P_4 + 5O_2 \rightarrow P_4O_{10}$	*Source: 7405/1 Jun 2023 Q03* Allow $4P + 5O_2 \rightarrow P_4O_{10}$	1
08.2	$SO_2 + H_2O \rightarrow H_2SO_3$	Allow $\rightarrow H^+ + HSO_3^-$ or $2H^+ + SO_3^{2-}$ Do not accept H_2SO_4	1
08.3	Displayed HSO_4^-: S with two S=O, one S–O^- and one S–O–H	Allow SO_4^{2-} displayed	1
08.4	M1 $Mg + H_2O \rightarrow MgO + H_2$ M2 white solid / white powder **or** (bright) white light / white flame	Do not accept white ppt or effervescence	2
08.5	$3MgO + 2H_3PO_4 \rightarrow Mg_3(PO_4)_2 + 3H_2O$		1